ADAR'S STUDY SERIES

LINEAR ALGEBRA

PART 3 · VECTORS, LINEAR COMBINATIONS & SPAN

Vector Operations · Linear Combinations · Span · Matrix-Vector Products · Existence of Solutions

Deep Violet & Amber Edition

CONTENTS

1 · OVERVIEW & GOALS

- **Learn what vectors are, and what R^n represents.**
- **Define a linear combination and the span of a set of vectors.**
- **Understand how to determine whether a vector $\vec{b}$ can be written as a linear combination of a set of vectors — both computationally and conceptually.**

2 · WHAT IS R^n?

$\mathbb{R}$— Real Numbers (1-Dimensional)

Represents the real number line.

```
   -2    -1     0     1     2
--|-----|-----|-----|-----|---
```

$\mathbb{R}^2$— The Plane

$$\mathbb{R}^2 = \{ \text{ all } (x, y) \text{ such that } x, y \in \mathbb{R} \}$$

(Set of ordered pairs)

$\mathbb{R}^3$— 3D Space

$$\mathbb{R}^3 = \{ \text{ all } (x, y, z) \text{ such that } x, y, z \in \mathbb{R} \}$$

(Set of ordered triples)

General Definition — $\mathbb{R}^n$

$$\mathbb{R}^n := \text{Set of ordered n-tuples } (x_1, x_2, \ldots, x_n) \text{ of reals}$$

(where ":=" means "defined to be equal to")

Note: Elements of $\mathbb{R}^n$ are usually written as a column vector:

R^n = set of all such columns, where $x_1, \ldots, x_n \in R$

$$\mathbb{R}^n = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad \text{where} \quad (x_1, x_2, \ldots, x_n) \in \mathbb{R}$$

3 · VECTOR NOTATION & THE ZERO VECTOR

General Vector

We write a general vector in $\mathbb{R}^n$ as:

$$\vec{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n$$

(The arrow above x represents that it is a vector.)

The Zero Vector

$$\vec{\mathbf{0}} = \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \\ \vdots \\ \mathbf{0} \end{bmatrix} \in \mathbb{R}^n$$

(The individual values inside a vector are called entries of the vector.)

4 · VECTOR OPERATIONS

We can add/subtract and scale vectors.

Let $\vec{x} = [x_1, \ldots, x_n]$, $\vec{y} = [y_1, \ldots, y_n] \in \mathbb{R}^n$, and let c ∈ ℝ be a scalar (constant).

Vector Addition

$$\vec{x} + \vec{y} := \begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \\ \vdots \\ x_n + y_n \end{bmatrix}$$

Vector Subtraction

$$\vec{x} - \vec{y} := \begin{bmatrix} x_1 - y_1 \\ x_2 - y_2 \\ \vdots \\ x_n - y_n \end{bmatrix}$$

Scalar Multiplication (Scaling)

$$c\vec{x} = \begin{bmatrix} c \cdot x_1 \\ \vdots \\ c \cdot x_n \end{bmatrix}$$

(Scalar) · (Vector) = Vector

5 · GEOMETRIC INTERPRETATION IN R²

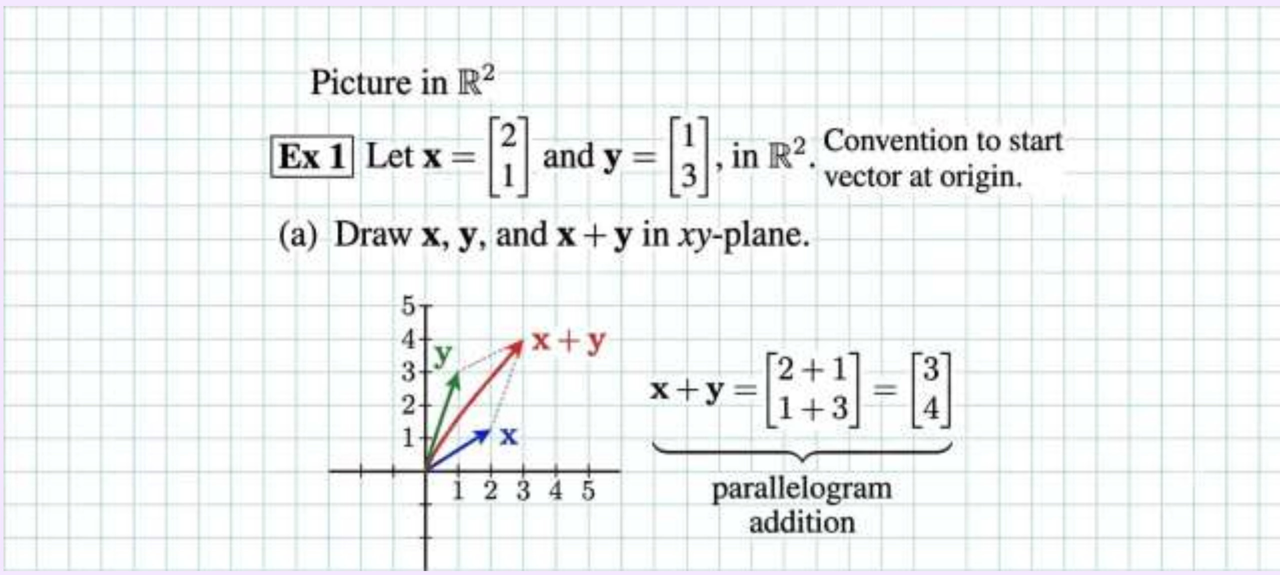
Convention: Always start vectors at the origin (0, 0).

Example 1 — Vector Addition

Let $\vec{x}$ = [2, 1] and $\vec{y}$ = [1, 3] $\in \mathbb{R}^2$.

$$\vec{x} + \vec{y} = [2+1,\ 1+3] = [3,\ 4]$$

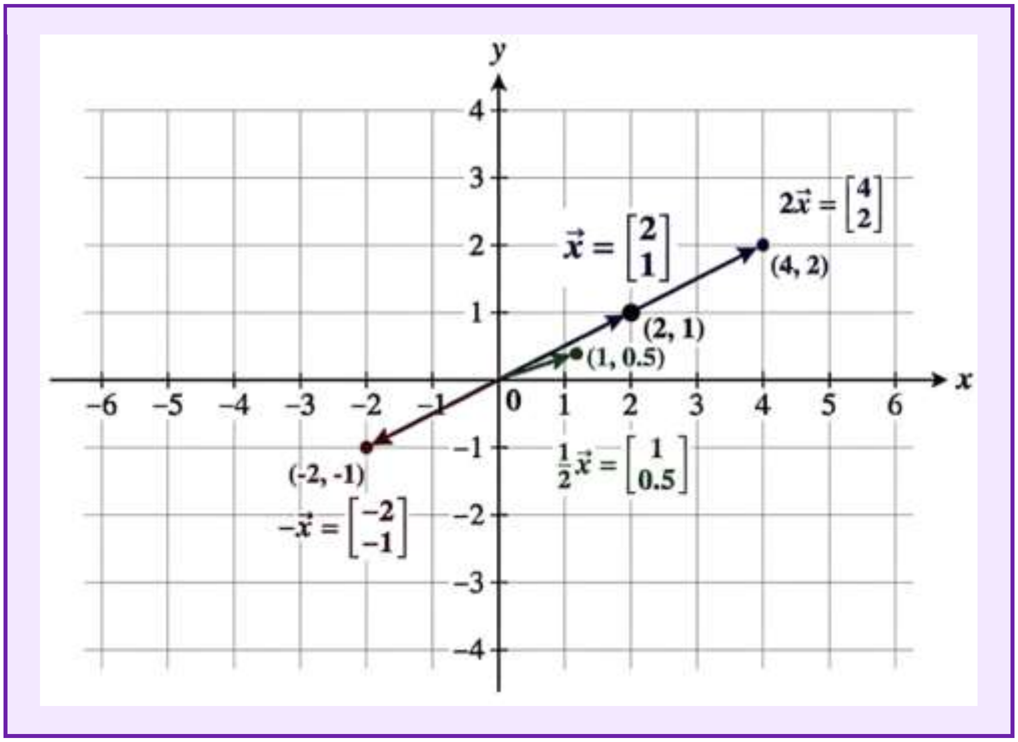
Parallelogram Rule for Addition



Example 2 — Scalar Scaling

Let $\vec{x}$ = [2, 1]. We sketch $\vec{x}$, $2\vec{x}$, $\frac{1}{2}\vec{x}$, and $-\vec{x}$:

- $2\,\vec{x}$ = [4, 2]
- $\frac{1}{2}\vec{x}$ = [1, 0.5]
- $-\vec{x}$ = [-2, -1]



6 · BASIC ARITHMETIC PROPERTIES OF VECTORS

Vectors in $\mathbb{R}^n$ satisfy fundamental algebraic properties for any vectors $\vec{u}, \vec{v}, \vec{w} \in \mathbb{R}^n$ and scalar c $\in \mathbb{R}$

Associativity of Addition

$$(\vec{u} + \vec{v}) + \vec{w} = \vec{u} + (\vec{v} + \vec{w})$$

Additive Identity

$$\vec{0} + \vec{v} = \vec{v}$$

Zero Scalar Multiplication

$$0 \cdot \vec{v} = \vec{0}$$

Distributivity

$$c(\vec{u} + \vec{v}) = c\,\vec{u} + c\,\vec{v}$$

7 · LINEAR COMBINATION & SPAN

Linear Combination

Definition: Let $\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_k$ be vectors in $\mathbb{R}^n$. Let c_1, c_2, ..., $c_k \in \mathbb{R}$ be scalars (called weights). Then the expression:

$\boldsymbol{c_1\vec{v}_1 + c_2\vec{v}_2 + \ldots + c_k\vec{v}_k}$ is called a linear combination of $\vec{v}_1$, $\vec{v}_2$, ..., $\vec{v}_k$.

Note: A linear combination of vectors results in another vector.

Span

Definition: The span of a set of vectors is the set of all possible linear combinations of those vectors:

Span$\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_k\}$ = All linear combinations of $\vec{v}_1$, ..., $\vec{v}_k$

Conceptual Meaning

Span$\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_k\}$ = Everywhere we can get to in $\mathbb{R}^n$ only traveling in direction of $\vec{v}_1$, ..., $\vec{v}_k$

Key Property

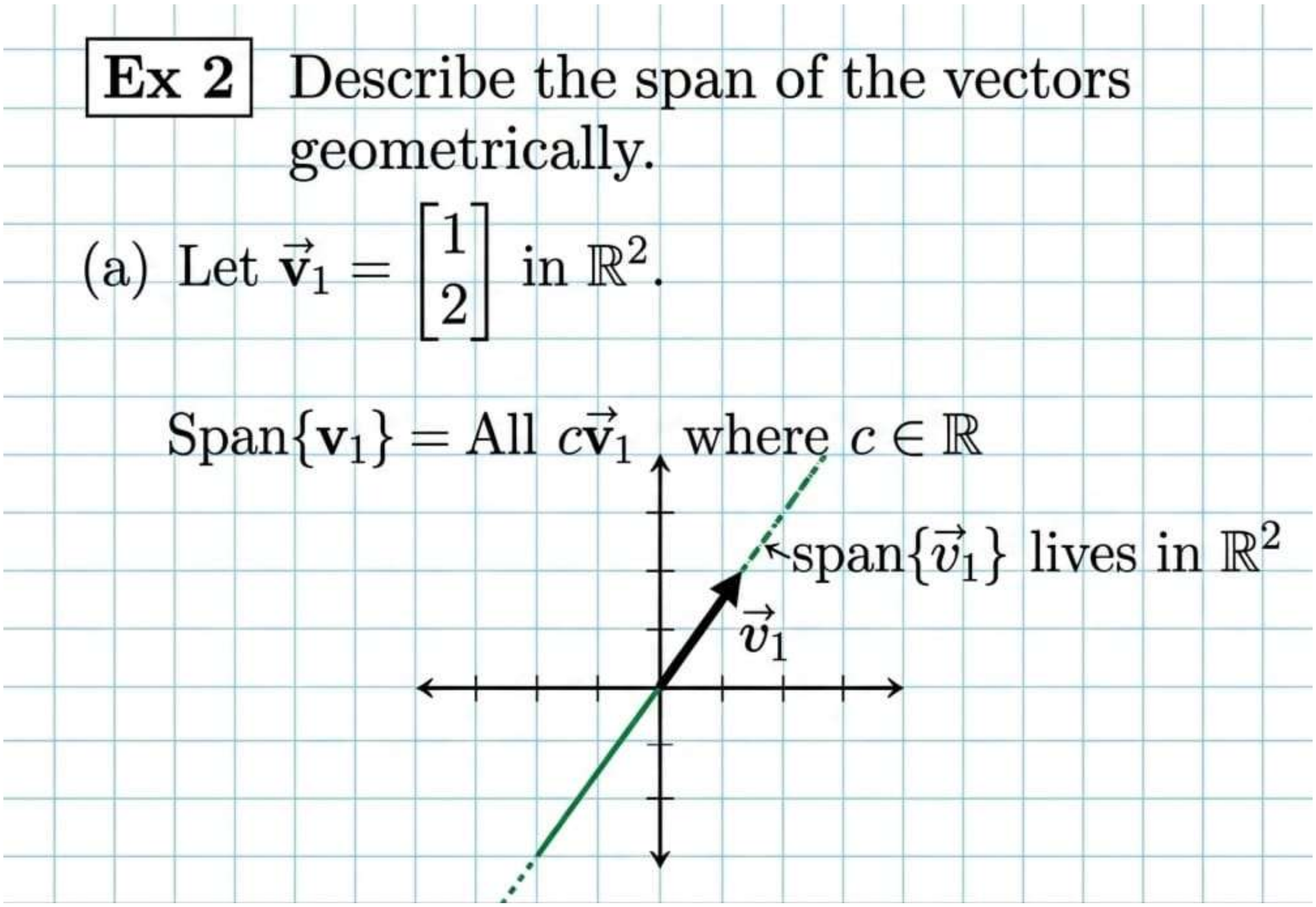
The zero vector $\vec{0}$ is always in Span $\{\vec{v}_1, \ldots, \vec{v}_k\}$.

Why? By choosing all weights to be 0, we get:

$$0\vec{v}_1 + 0\vec{v}_2 + \ldots + 0\vec{v}_k = \vec{0}$$

8 · GEOMETRIC EXAMPLES OF SPAN

Example 2(a) — Single Vector in $\mathbb{R}^2$



Example 2(b) — Two Linearly Independent Vectors in $\mathbb{R}^3$

Let $\vec{v}_1 = [1, 0, 0]$ and $\vec{v}_2 = [0, 1, 0]$ in $\mathbb{R}^3$.

$$c_1\vec{v}_1 + c_2\vec{v}_2 = c_1[1,0,0] + c_2[0,1,0] = [c_1, c_2, 0]$$

Geometric Interpretation: The xy-plane in 3D-space ($\mathbb{R}^3$).

Important Note: Span$\{\vec{v}_1, \vec{v}_2\} \neq \mathbb{R}^2$ because the vectors in this span have 3 components and live in $\mathbb{R}^3$, whereas elements of $\mathbb{R}^2$ only have 2 components.

Example 2(c) — Two Parallel (Dependent) Vectors in $\mathbb{R}^3$

Let $\vec{v}_1$ = [1, 3, -4] and $\vec{v}_2$ = [$\frac{1}{4}$, $\frac{3}{4}$, -1] in $\mathbb{R}^3$.

Observe that $\vec{v}_1 = 4\vec{v}_2$. Substituting this into a general linear combination gives:

$$c_1\vec{v}_1 + c_2\vec{v}_2 = c_1(4\vec{v}_2) + c_2\vec{v}_2 = (4c_1 + c_2)\,\vec{v}_2$$

Since $(4c_1 + c_2)$ is just a scalar constant:

Span{$\vec{v}_1$, $\vec{v}_2$} = Span{$\vec{v}_2$} = Line through origin and [$\frac{1}{4}$, $\frac{3}{4}$, -1] in $\mathbb{R}^3$

9 · MATRIX REPRESENTATION OF A LINEAR SYSTEM

Suppose a linear system has m equations and n variables. Its augmented matrix is expressed as:

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} & | & b_1 \\ a_{21} & a_{22} & \dots & a_{2n} & | & b_2 \\ \vdots & \vdots & & \vdots & | & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} & | & b_m \end{bmatrix}$$

$$= [\ \vec{a}_1 \quad \vec{a}_2 \ \dots \ \vec{a}_n \ | \ \vec{b}\ ]$$
$$= [\ A \ | \ \vec{b}\ ]$$

- **A = [$\vec{a}_1, \vec{a}_2, \ldots, \vec{a}_n$] is the m × n coefficient matrix.**
- **Each column $\vec{a}_1, \vec{a}_2, \ldots, \vec{a}_n$ is a column vector in $\mathbb{R}^m$.**
- **$\vec{b}$ = [$b_1, b_2, \ldots, b_1$] is the right-hand side constant vector in $\mathbb{R}^m$.**

10 · EQUIVALENCES FOR SYSTEM SOLUTIONS

Let $\vec{x} = [x_1, x_2, ..., x_n] \in \mathbb{R}^n$ be a candidate solution vector for [A | $\vec{b}$]. The following statements are equivalent:

1. System of Equations Representation

$$
\begin{aligned}
a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= b_1 \\
a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= b_2 \\
&\vdots \\
a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &= b_m
\end{aligned}
$$

2. Vector Equation Representation

$$
x_1 \cdot \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix} + x_2 \cdot \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix} + \dots + x_n \cdot \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}
$$

$$
x_1\vec{a}_1 + x_2\vec{a}_2 + \dots + x_n\vec{a}_n = \vec{b}
$$

3. Linear Combination / Span Interpretation

$\vec{b}$ is a linear combination of column vectors $\vec{a}_1, ..., \vec{a}_n$.

Equivalently: $\vec{b} \in \text{Span}\{\vec{a}_1, ..., \vec{a}_n\}$

11 · THE MATRIX-VECTOR PRODUCT $A\vec{x}$

Definition: If A = $[\vec{a}_1\ \vec{a}_2\ \dots\ \vec{a}_n]$ is an m × n matrix (with m rows and n columns, where $\vec{a}_i \in \mathbb{R}^m$) and $\vec{x} = [x_1, x_2, \dots, x_n] \in \mathbb{R}^n$, then the product $A\vec{x}$ is defined as:

$$A\vec{x} := x_1\vec{a}_1 + x_2\vec{a}_2 + \dots + x_n\vec{a}_n$$

This product $A\vec{x}$ represents a linear combination of the columns of A with weights given by the entries of $\vec{x}$, resulting in a vector in $\mathbb{R}^m$.

12 · FUNDAMENTAL THEOREM — EXISTENCE OF SOLUTIONS

The following statements are equivalent:

The linear system $[A \mid \vec{b}]$ has a solution

$\Leftrightarrow$ $\vec{b}$ is a linear combination of $\vec{a}_1, \dots, \vec{a}_n$ (same as saying $\vec{b} \in \text{span}\{\vec{a}_1, \dots, \vec{a}_n\}$)

$\Leftrightarrow$ $A\vec{x} = \vec{b}$ for some $\vec{x} \in \mathbb{R}^n$

$\Leftrightarrow$ The echelon form of $[A \mid \vec{b}]$ has NO row of the form $[0 \dots 0 \mid \blacksquare]$ with $\blacksquare \neq 0$

13 · WORKED EXAMPLE — TESTING SPAN MEMBERSHIP

Example 3

Question:

`Is [1, 2, 3] ∈ Span{ [-1,1,2], [2,1,0], [1,2,2] } ?`

(i.e., is [1, 2, 3] a linear combination of those vectors?)

Solution Procedure

Check if the linear system given by the augmented matrix is consistent:

Initial Matrix

```
[ -1   2   1 |  1 ]
[  1   1   2 |  2 ]
[  2   0   2 |  3 ]
```

Apply row operations: $R_1 + R_2 \rightarrow R_2,\ 2R_1 + R_3 \rightarrow R_3$

```
[ -1   2   1 |  1 ]
[  0   3   3 |  3 ]
[  0   4   4 |  5 ]
```

Simplify rows: (-1) $R_1 \rightarrow R_1$, (1/3) $R_2 \rightarrow R_2$

```
[  1  -2  -1 | -1 ]
[  0   1   1 |  1 ]
[  0   4   4 |  5 ]
```

Eliminate Row 3: $-4R_2 + R_3 \rightarrow R_3$

```
[  1  -2  -1 | -1 ]
[  0   1   1 |  1 ]
[  0   0   0 |  1 ]
```

Conclusion

The row [0 0 0 | 1] indicates that the system is inconsistent.

```
[1, 2, 3]  is NOT in Span{ [-1,1,2], [2,1,0],
[1,2,2] }
```

14 · WORKED EXAMPLE — FINDING WEIGHTS IN A SPAN

Example 4

Question:

$$\text{Is } \vec{b} = [-9, 3, 10] \in \text{Span}\{ [-1,1,2], [2,1,0] \} \text{ ?}$$

(If yes, find weights x_1, x_2 such that $\vec{b} = x_1\vec{v}_1 + x_2\vec{v}_2$)

Solution Procedure

Set up the augmented matrix:

$$\left[\begin{array}{cc|c} -1 & 2 & -9 \\ 1 & 1 & 3 \\ 2 & 0 & 10 \end{array}\right]$$

Row operations to Echelon Form: $R_1 + R_2 \to R_2$, $2R_1 + R_3 \to R_3$

$$\left[\begin{array}{cc|c} -1 & 2 & -9 \\ 0 & 3 & -6 \\ 0 & 4 & -8 \end{array}\right]$$

Rescale rows: $R_1(-1) \to R_1$, $R_2(1/3) \to R_2$, $R_3(1/4) \to R_3$

$$\left[\begin{array}{cc|c} 1 & -2 & 9 \\ 0 & 1 & -2 \\ 0 & 1 & -2 \end{array}\right]$$

Eliminate Row 3: $R_2 - R_3 \rightarrow R_3$

Echelon Form (EF)

$$\left[\begin{array}{cc|c} 1 & -2 & 9 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{array}\right]$$

Observation: The Echelon Form has no row of the form [0 … 0 | ■] where ■ ≠ 0. Thus, a solution exists!

Back Substitution / RREF: $2R_2 + R_1 \rightarrow R_1$

Reduced Row Echelon Form (RREF)

$$\left[\begin{array}{cc|c} 1 & 0 & 5 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{array}\right]$$

Conclusion

$$x_1 = 5$$
$$x_2 = -2$$

Thus:

$$[-9, 3, 10] = 5\cdot[-1, 1, 2] - 2\cdot[2, 1, 0]$$

15 · FUNDAMENTAL THEOREM — SPANNING THE ENTIRE SPACE R^m

Note: Key Equivalent Conditions

Let A be an m × n matrix with columns $\vec{a}_1, \ldots, \vec{a}_n$. The following statements are equivalent (⟺):

1. **Linear system $[A \mid \vec{b}]$ has a solution for every $\vec{b} \in \mathbb{R}^m$.**
2. **Span{ $\vec{a}_1, \ldots, \vec{a}_n$ } = $\mathbb{R}^m$**
3. **The reduced matrix never has a row of the form [0 … 0 | ■] (where ■ ≠ 0).**
4. **The coefficient matrix A has a pivot in every row.**

16 · WORKED EXAMPLE — TESTING FULL SPACE SPAN

Example 5

Question:

`Do vectors [1,2,3], [1,0,1], [2,-1,1] span all of ℝ³ ?`

Solution Procedure

Form the matrix A and check if there is a pivot in each row:

```
    [ 1   1   2 ]
A = [ 2   0  -1 ]
    [ 3   1   1 ]
```

Row Reduction

Initial Coefficient Matrix:

```
[ 1   1   2 ]
[ 2   0  -1 ]
[ 3   1   1 ]
```

Eliminate lower entries in Column 1: $-2R_1 + R_2 \rightarrow R_2$, $-3R_1 + R_3 \rightarrow R_3$

```
[ 1   1   2 ]
[ 0  -2  -5 ]
[ 0  -2  -5 ]
```

Eliminate Row 3: $R_3 - R_2 \rightarrow R_3$

```
[ 1   1   2 ]
[ 0  -2  -5 ]
[ 0   0   0 ]
```

Conclusion

Since row 3 contains all zeros, not every row has a pivot (there are only 2 pivots, but 3 rows are required).

The columns do NOT span $\mathbb{R}^3$.

17 · KEY POINTS & SUMMARY

- **$\mathbb{R}^n$ is the set of ordered n-tuples, usually written as column vectors.**
- **Vectors can be added, subtracted, and scaled entry wise.**
- **Vector addition follows the parallelogram rule geometrically.**
- **A linear combination $c_1\vec{v}_1 + \ldots + c_k\vec{v}_k$ always produces another vector.**
- **Span$\{\vec{v}_1, \ldots, \vec{v}_k\}$ is the set of ALL linear combinations — and always contains $\vec{0}$.**
- **A linear system $[A \mid \vec{b}]$ has a solution $\iff$ $\vec{b} \in$ Span {columns of A} $\iff$ $A\vec{x} = \vec{b}$ for some $\vec{x}$.**
- **A pivot-free row in the constants column of the echelon form guarantees a solution exists.**
- **Columns of A span the ENTIRE space $\mathbb{R}^m$ only when A has a pivot in every row.**

END OF PART 3 · NEXT: PART 4 — MATRIX EQUATION & PARAMETRIC VECTOR FORM